

Project APEX - Documentation & User Flow

Executive Summary:

Project APEX or APEX trading terminal is a fully automated AI powered option trading platform, which is completely hassle free for a trader as they only need to enter the ticker of any U.S. based company, select the chart-timeframe and adjust the view of the interactive technical chart, select the expiry date options from the dropdown, and then hit SCAN. After hitting scan the APEX AI will run a deep synthesized analysis consisting of:

- A 10 layer deep technical analysis(MACD, EMA, RSI, BOLLINGER BANDS, PIVOT POINTS, SUPPORT/RESISTANCE, SUPERTREND, VOLUME, TECHNICAL CANDLESTICK PATTERNS)
- Upon running a 10 layer deep technical analysis on the charts the AI will start analyzing the entire option chain based on the user selected expiry dates. This part of the analysis mainly checks and looks at the greeks (DELTA, THETA, GAMMA, VEGA) as well as the bid/ask spread, volume, change, and open interest(OI) to locate a proper set up.
- Next it will compare implied volatility (IV) with historical volatility(HV) to determine the ideal scenario of buying or selling the contract.
- Next the system will scan and analyze for live real time sentimental thesis from financial news and other reliable sources as well the technical charts and option charts to understand investor sentiment.
- The system will also pull a fundamental snapshot of the company such as the earnings per share(EPS), revenue, price/earnings ratio(P/E), sales, income, analysts estimates, growth rates/expectation to run a scan that will determine the health of the company.
- After collecting all the information mentioned above the individual layers of information get synthesized from all the layers and runs a deep analysis and research model that will give a rating/score.

- Based on the rating it gives a simple brief plain english explanation and gives the exact strategy, methodology, and execution steps to be followed.
- The terminal is instructed to automatically execute and follow the strategy if it crosses a certain rating/score of (85).
- Lastly, it gives a final properly formatted certificate looking that contains the order details and conformation.

APEX trading is not a dashboard that just gathers and displays data, rather it is a full cycle AI powered trading agent/terminal. It logs, analyzes, and executes every order with analytical backing and discipline.

The main innovation of the APEX is the “**APEX STRATEGY**”.

We have designed this strategy to capitalize on the volatility spike before a major catalyst event and the volatility crush that follows it.

The strategy works in 2 sorts of mechanics one being where you buy and one where you sell the option.

At first you are given a major upcoming event related to a specific company or the broader market, while leading up to that event you are going to do the 2 mechanics we discussed for the strategy to work in the intended manner.

While leading up to the event you buy a call option and a put option both OTM and both expiring a week after the event.

- You want to buy OTM as it is much cheaper, with the call strike above and the put strike below the current price by about the expected move. After the event these long options keep more value than the short ones, because back-week IV falls less than front-week.
- The time frame should be about 1 week after the event so the long options still hold time value once the short options expire the day after the event.

This marks one of the two steps of the strategy, the second step is when you sell both an OTM call option and put option to their respective strike prices as mentioned in the first step and they should have an expiry date of immediately the next day of the event.

Now, in this step you want to sell options because as you are leading up to the event the front-week IV is significantly higher than HV and back-week IV, so you can squeeze out a higher premium when selling.

- You want to sell OTM at the same strikes as the long options. The trade profits most when the stock lands near one of the strikes. If the stock stays flat, both short options expire worthless but the long options lose value too, so expect a moderate loss. A very large move also loses. The most you can lose is the net debit paid.
- The expiry should be immediately the next day because Theta(Time Decay) and the IV crush hit the short options hardest after the event.

Section 1 - What problem are we solving:

The main problem that modern option traders are facing is the lack of hassle free and all in one place analysis. Thus, why APEX is the one stop solution that will take care of these major issues such professionals face:

- **Analysis:**
 - There are many benchmarks and tools that traders need to scan and analyze, but all the software that provides these details are scattered.
- **Volatility Misinterpretation:**
 - Misreading volatility charts such as entering a long position when IV high or doing the opposite can cause traders to face severe losses.

- **Greeks:**
 - Blindly opening positions without reading, understanding, and analyzing the greeks can sometimes make you pay a hefty fine.
- **Faulty Execution:**
 - Having one trade where the execution works perfectly is not the case for every trade set up and it is important to have a systematic and streamlined approach.
- **Strategy Mismatch:**
 - Not knowing what strategy should be used in a situation or using a strategy without full understanding of the strategy or even using the wrong strategy can cause mishappenings.

APEX ensures full accuracy and credibility as it goes through each scenario, understands, and analyzes the details and gives evidence backing to prevent professionals from facing such major issues.

Section 2 - The Framework:

For APEX to work in the intended manner there is no 1 metric that has to pass: we have a 5 layer deep analysis:

1. **TREND** → Super Trend + EMA (9,21,50,100/200)
2. **MOMENTUM** → RSI (IM) (over bought) - MACD (crossover + histogram)
3. **VOLATILITY** → Ballinger Band Span width + IV vs. HV & IV Rank.
4. **PATTERNS** → Candlestick chart recognition
5. **PRICE LEVEL** → Pivot Points + Support / resistance
6. **GREEKS** → Delta, Treta, Gamma, & i vega to get price perfection.
7. **OPTION CHAIN** → OI change. Volume Change, Bia/ask Spread. Change, Strike. Expiry, IV. Call put.
8. **SENTIMENTAL** → News articles, filings, direction flow, etc.
9. **FUNDAMENTAL** → EPS, P/E, Revenue, Sales. Income, Profit, Growth, Analyst Expectations
10. **FINAL STRATEGY** → fully Comprehensive final synthesis of research.

Section 3 - The UX Walkthrough

As the user opens the APEX terminal and lands on the login screen they need to enter their username, password, and a six digit unique key that changes every time the market opens.

Upon logging in they can see the main dashboard which has a big technical chart of the S&P500 and that chart contains all the key technical indicators such as: (MACD, EMA, RSI, BOLLINGER BANDS, PIVOT POINTS, SUPERTREND, VOLUME) and the chart allows you to add in trend lines for drawing and identifying key levels. The main dashboard will also have a watchlist monitoring section and the ability to track any open positions. It also has the sentiment section to read, analyze, and understand sentimental news.

Above the main chart there is an entry field where you can enter the ticker to any US based company, select the time frame of the chart, adjust the chart view, and choose the expiry date based on the dropdowns. After all that the user just needs to click scan and the APEX terminal will run its 5 layer deep analysis magic.

Next it loads up a collapsable option chain and technical graph that shows the details.

On the left side of the terminal there will be basic stock information that will be displaying the company name, current OHLCV price data, and other important and needed benchmark metrics.

Section 4 - Technical Analysis:

The first that is scanned & analyzed by APEX terminal are the candlestick patterns on the technical charts as they give ideas on trend, target, stop loss, entry, & exit.

Bullish Reversals (Buy at Support Signal) → Hammer. Bullish engulfing. Morning Star, dragonfly doji. piercing line, inverted hammer. bullish harami)

Bearish Reversal (Sell at resistance signal) → Shooting Star. Bearish engulfing, evening Star. Dark Cloud Cover, hanging man. bearish harami)

Trend Continuation (Same trend) → Three-white soldiers, Three-black Crows • Rising three methods

Indecision (Further Conclusion required) → Spinning top, Standard daji. Inside bar

These categories & pattern sets give ideas about trend & future price expectations ONLY they confirm trend & price action NOT breakouts/breakdowns.

Bullish	Bearish
Flag (Bullish)	Flag (Bearish)
Ascending Triangle	Descending Triangle
Cup & Handle	Inverted Cup & Handle
Inverted Head & Shoulder	Head & Shoulder
Double Bottom	Double Top
Rounding Bottom	Rounding Top

These are just Standard candlestick patterns that confirm trend, price action & expectation establishment. & breakout/breakdown. These are paired with more indicators for better analysis:

MACD Moving Average - Convergence-Divergence)

- Standard Configuration: 12 period EMA. 26 period EMA. 9 period signal line.
- Bullish:
 - When the MACD line crosses the signal line from below or MACD line is above zero line.
- Bearish:

- When the MACD line crosses the signal line from above or MACD line is below zero line.

Bullish Divergence

When price movement (candlesticks) are making a lower low pattern while MACD is making higher lows, signaling seller exhaustion & bullish reversal.

Bearish Divergence

When price movement (candlesticks) are creating higher highs patterns while MACD is making lower highs, signaling buyer exhaustion & bearish reversal.

RSI (Relative Strength Index) (14 TRADING PERIODS)**Bullish**

RSI level above & sustaining above 60-70 & crossing 70 means it has entered strong upward momentum & can stay high for a long period. RSI levels of 40-50 is supported during bull run & a classic strong entry point would be when RSI dips below 30 & instantly crosses back above 30, while sustaining it.

Bearish

RSI level below & sustaining below 40-30 & crossing the levels of 30 means it has entered a strong bearish trend & can stay low for a consistent period. RSI levels of 50-60 is resistance during bearish trend & a classic sell/short signal occurs when RSI crosses 70 & dips right below 70, while sustaining it.

Bullish Divergence

When price action (candlestick) is making a lower low pattern but RSI is displaying a higher low, signaling bearish exhaustion & bullish reversal.

Bearish Divergence

When price action (candlestick) is making a higher high pattern but RSI is displaying a lower high, signaling buyer exhaustion & showing signs of bearish reversal.

EMA (Exponential Moving Averages)

For technical analysis, trend confirmation, & price action set up we focus on EMA Stacks of (9, 21, 50, 100, 200) ranging from both immediate short term to mid-long term.

Bullish

When the price is constantly above & sustaining above each level of the EMA Stack starting from short/immediate (9) to mid/long (200) showing a consistent & strong uptrend. The best & classic entry point is when it touches the 9-21 EMA & sustaining above them as well as crosses those levels. The crossover of a shorter term moving average crosses the longer term from the bottom, kind of like MACD.

- Golden Crossover (50 EMA crosses 200 EMA from bottom)
- When the price is unusually below the moving averages it often sets up an upward pullback

Bearish

When the price is constantly below the EMA Stack, again starting from short/immediate (9) to mid/long (200). The classic & strongest entry point would be the pullback entry — when the price goes below the EMA, rises back near to touch the EMA line, and once it is near that line look for reversal signals in candlestick patterns & enter bearish trade. The crossover of a short term moving average to long term average from above.

- Death Crossover (50 EMA crosses 200 EMA from above)
- When price is unusually above EMA it suggests a reversal

The EMA number changes based on the timeframe used by the trader. For intraday & day trading the 9-21 EMAs are most applicable, for standard swing trading 12-26 EMA, for longer swings the 50-200 EMA. On a little longer term vision the 50-200 EMAs are highly effective.

Supertrend

The Supertrend is a trend-following indicator calculated using ATR (Average True Range) to account for market volatility.

Bullish

The candlestick crosses the current red line switching the trend to green, marking the start of an uptrend & the price where the trend switches would act as support/stop-loss.

Bearish

When the candlestick goes below the green-line price line it marks a potential trend reversal & that price line becomes a resistance.

Bollinger Bands

The Bollinger Bands consist of 3 lines:

- A middle simple moving average (20 periods)
- The other 2 are upper resistance level & lower support level which is a standard deviation of usually 2 away from middle

Bullish

When the price action (candlestick) crosses the middle line, sustains above it and breaks out by crossing the upper band & breaking out suggesting a potential uptrend. The breakout price level acts as a support line.

Bearish

When the price action (candlestick) goes below the midline, sustains below it and breaks down below by crossing the lower band suggesting a downtrend. The breakdown level acts as the resistance level.

Bullish (W-Pattern)

The price action (candlestick) breaks below the midline & slides near or even beyond the lower bound but it retreats back to the midline & again pulls back to the earlier support level & then shows an uptrend again by breaking out above midline making it an ideal entry point.

Bearish (M-Pattern)

The price action (candlestick) breaks above the upper band or gets near that level & pulls back to the midline, gets a support at the midline & bounces back to the earlier resistance to again pull back & this time break the midline showing signs of a potential downtrend with the midline being the resistance — marking the breaking of the midline a perfect entry to enter a bearish trade.

Pivot Points

Standard Formula: $(\text{High} + \text{Low} + \text{Close}) / 3$

- $R1: (PP \times 2) - \text{Low}$; $R2: PP + (\text{High} - \text{Low})$; $R3: \text{High} + 2 \times (PP - \text{Low})$
- $S1: (PP \times 2) - \text{High}$; $S2: PP - (\text{High} - \text{Low})$; $S3: \text{Low} - 2 \times (\text{High} - PP)$

Pivot points are historical horizontal price levels calculated from previous day's highs & lows & close, projecting key future levels.

Bullish

When the price is above the central pivot level the upward trend is continued with the support level being at pivot & targets expanding outward to R1, R2, & R3.

Bearish

When the price is below the central pivot level the downward trend is continued with resistance level being at pivot & downward targets expanding to S1, S2, & S3.

Support & Resistance

These are structural price levels where the stock price has historically stopped or reversed from, which based on supply & demand scenarios.

Bullish

When the stock price breaks above the resistance level (price level where trend reverses) suggesting a potential upside.

Bearish

When the stock price breaks below the support level (price level stock price doesn't break) suggesting a bearish trend.

When the support or resistance is broken but the price again retreats to those levels in short span it shows a false breakout & potentially more buying/selling pressure.

Volume

Volume represents the number of shares, contracts, or tokens of an underlying asset that are traded on a daily basis.

Bullish

When the price is constantly in a bullish trend making higher highs & the volume is also increasing but if the stock sees a dip so does the volume, indicating that there is a strong buyer demand.

Bearish

When the price is constantly in a bearish run making lower lows & the volume decides to follow & if the stock trend reverses so does volume but not by a huge amount showing strong seller environment.

Bullish Divergence

The price is making lower lows but volume is making higher lows showing high demand & buying at low levels suggesting reversal.

Bearish Divergence

The price is making higher highs but volume is at lower highs showing selling pressure suggesting reversal from current trend.